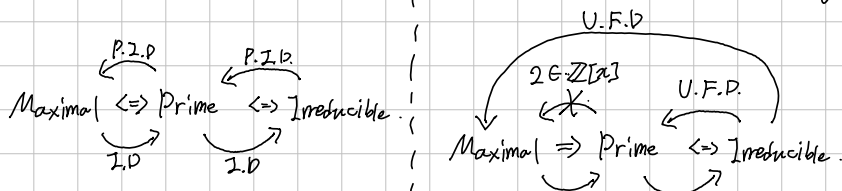
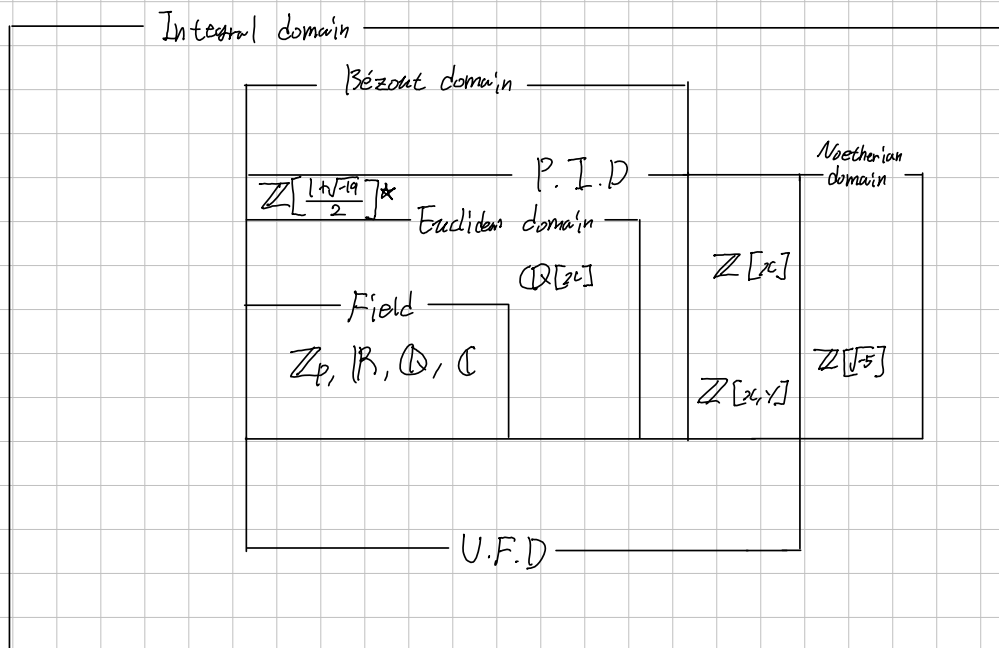


Field  $\Rightarrow$  Euclidean Domain  $\Rightarrow$  Principal Ideal Domain  $\Rightarrow$  Unique Factorization Domain  $\subset$  Integral Domain.



P.I.D.  $\Leftarrow$  Noether  $\Leftrightarrow$  Every Ideal are finitely generated  $\Leftrightarrow A.C.C.I \Rightarrow A.C.C.P \Rightarrow \exists$  factorization  $\Leftarrow$  U.F.D.

Bézout  $\Leftrightarrow$  Finitely generated ideal is principal  $\Rightarrow$  irreducible = prime  $\Rightarrow$  ! factorization  $\Leftarrow$  U.F.D.



Basic languages for generated ideal.

Def. Let  $R$  be a Commutative Ring, and  $a, b \in R$  with  $b \neq 0$ .

1.  $a$  is called a multiple of  $b$  if: there exists  $x \in R$  s.t.  $a = bx$ . Simply  $b|a$ .
2. A non-zero  $d \in R$  is called a Greatest Common divisor of  $a$  and  $b$  if;

$$d|a \text{ and } d|b \text{ (or, } a, b \in (d)), \quad d'|a \text{ and } d'|b \Rightarrow d'|d. \\ (a, b \in (d') \Rightarrow (d) \subseteq (d'))$$

Note:  $b|a \Leftrightarrow a \in (b) \Leftrightarrow (a) \subseteq (b)$ .

Directly,  $(a, b) = (d)$  implies  $d$  is the g.c.d of  $a$  and  $b$ ,  
but Conversely, gcd maybe does not generate the ideal; in  $\mathbb{Z}[x]$ ,

$$(1) \neq (2, x) \text{ but } 1 \text{ is gcd of } 2 \text{ and } x$$

Prop. Let  $R$  be a Commutative Ring, and  $a, b, c \in R$  be given. Then,

- 1).  $(a, b) = (a) + (b)$
- 2).  $(c) \cdot [(a) + (b)] = (c) \cdot (a) + (c) \cdot (b) = (ca) + (cb)$ .
- 3).

